

Add Employee — Gold Standard

The **single-requirement gold standard** for Phase 1. This is the bar: a senior QA architect cannot easily think of an important scenario that LevelUp missed.

Not a leaderboard. Not a score to optimise. It is the exhaustive list of scenarios an experienced QA architect would expect a test suite for **Add Employee** to contain, so we can generate, compare, and fix whatever is missing — one capability at a time.

The requirement under test

Add Employee (HR / Employee Management)

An authorized admin adds a new employee by entering **first name**, **last name**, an **employee ID**, and an **optional profile photo**. The employee ID may be **auto-generated or entered manually** and **must be unique**. On save the record is created, a **success notification** is shown, the admin is **redirected to the employee list**, and the new employee becomes **searchable by ID and by name**.

How to read this file

- Every scenario a senior QA would expect is listed under its category.
 - **Priority** — `Critical` (a defect here is a serious escape) → `High` → `Medium` → `Low` (polish / nice-to-have).
 - **Detection keywords** — how the audit harness (`scripts/add-employee-audit.ts`) decides the generated suite covers this scenario. Any one phrase present in a generated scenario's title/objective/risk-area (token-stemmed, via the shared `coverage-match` matcher) counts as covered. These describe the concept — they are **not** a script to match verbatim.
 - **Grounding** — `explicit` = the requirement states it directly; `expected` = a senior QA expects it for an HRM Add-Employee form even though the one-paragraph requirement does not spell it out. A miss on an `expected` scenario is a real conversation ("should the generator raise this?"), not automatically a defect.
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1. Functional — core creation paths

ID	Scenario	Priority	Grounding	Detection keywords
FUN-01	Create employee with all fields valid (incl. photo)	Critical	explicit	create employee, valid employee, add employee, create a record with valid
FUN-02	Create with mandatory fields only (photo omitted — it is optional)	High	explicit	minimum required, mandatory fields only, required fields only, without a photo, optional photo omitted
FUN-03	Create with an auto-generated employee ID	High	explicit	auto-generated id, auto generated id, auto-generated id
FUN-04	Create with a manually-entered employee ID	High	explicit	manually entered id, manual id, enter id manually
FUN-05	Created employee is retrievable / appears in the employee list	High	explicit	appears in the employee list, retrievable, record is retrievable, view an existing record

2. Validation — field-level, one scenario per field

ID	Scenario	Priority	Grounding	Detection keywords
VAL-01	First name blank is rejected	Critical	explicit	first name blank , first name required , missing first name , first name empty
VAL-02	Last name blank is rejected	Critical	explicit	last name blank , last name required , missing last name , last name empty
VAL-03	Both names blank is rejected	High	explicit	both names blank , both fields blank , all required fields blank
VAL-04	Employee ID blank (manual entry) is rejected	High	explicit	employee id blank , id required , missing employee id , id empty
VAL-05	First name whitespace-only is rejected (trimmed to blank)	High	expected	first name whitespace , whitespace-only first name
VAL-06	Last name whitespace-only is rejected (trimmed to blank)	High	expected	last name whitespace , whitespace-only last name
VAL-07	Invalid employee ID format is rejected	High	explicit	invalid id format , malformed id , invalid employee id format

ID	Scenario	Priority	Grounding	Detection keywords
VAL-08	Numeric / digit characters in a name are handled per rule	Medium	expected	numbers in name , numeric name , digits in first name
VAL-09	Field-level validation errors are shown inline	Medium	expected	inline validation error , field-level error , validation message shown

3. Business Rules — the logic that guards the data

ID	Scenario	Priority	Grounding	Detection keywords
BIZ-01	Duplicate employee ID is rejected (uniqueness)	Critical	explicit	duplicate , already exists , unique constraint , existing id
BIZ-02	Auto-generated ID is unique	High	explicit	auto-generated id is unique , generated id unique , unique auto
BIZ-03	Manually-entered ID accepted when unique	Medium	explicit	manual id accepted , unique manual id
BIZ-04	Employee ID uniqueness enforced case-insensitively	Low	expected	case-insensitive id , case insensitive unique
BIZ-05	Re-submitting the same form does not create a duplicate record	High	expected	double submit , duplicate submission , resubmit , idempotent create

4. Boundary — at, below and above every limit

ID	Scenario	Priority	Grounding	Detection keywords
BND-01	First name at maximum length is accepted	High	expected	first name max length , max length name accepted , name at maximum length
BND-02	First name one character over maximum is rejected	High	expected	name exceeds max length , name too long , over maximum length
BND-03	Employee ID at maximum length is accepted	Medium	expected	id max length , employee id maximum length
BND-04	Employee ID exceeding maximum length is rejected	High	expected	id too long , id exceeds max length , employee id over maximum
BND-05	Single-character name is accepted (minimum boundary)	Medium	expected	single character name , minimum length name , one character name
BND-06	Leading zeros in employee ID are preserved	Medium	expected	leading zero
BND-07	Unicode / accented characters in names are accepted (José, O'Brien)	Medium	expected	unicode name , accented name , special characters in name , non-ascii name
BND-08	Emoji in a name field is handled per rule	Low	expected	emoji

5. Security — hostile input must never win

ID	Scenario	Priority	Grounding	Detection keywords
SEC-01	SQL-injection string in a name field is neutralised	Critical	expected	sql injection , injection
SEC-02	XSS / script payload in a name field is escaped (not executed)	Critical	expected	xss , script injection , cross-site scripting , script payload
SEC-03	SQL-injection string in the employee ID is neutralised	High	expected	injection in id , sql injection id
SEC-04	HTML injection in a field is sanitised on display	Medium	expected	html injection , markup injection
SEC-05	An executable disguised as an image is rejected on upload	High	expected	spoofed , disguised , executable disguised , content type mismatch

6. Authorization — only the right user, only the right way

ID	Scenario	Priority	Grounding	Detection keywords
AUZ-01	Authorized admin can create an employee	High	explicit	authorized , admin can create , permitted user
AUZ-02	Unauthorized / non-admin user is blocked	Critical	explicit	unauthorized , forbidden , access denied , no permission , not allowed
AUZ-03	Direct URL / API access to the create endpoint is authorization-checked	High	expected	direct url , direct api , endpoint access control , bypass the ui
AUZ-04	Unauthenticated user is redirected to login	Medium	expected	unauthenticated , redirected to login , not logged in

7. Upload — the optional profile photo

ID	Scenario	Priority	Grounding	Detection keywords
UPL-01	Valid image (jpg/png) uploads and is associated with the record	Medium	explicit	valid file upload , valid photo , valid image , file upload is accepted
UPL-02	Save succeeds with no photo (photo is optional)	High	explicit	without a photo , photo is optional , no photo , optional photo omitted
UPL-03	Unsupported file type is rejected	High	expected	invalid file type , unsupported format , wrong format , disallowed type
UPL-04	Oversized image is rejected at the size boundary	High	expected	oversized , large file , file too large , exceeds max size
UPL-05	Image at exactly the maximum size is accepted	Medium	expected	at the limit , at maximum size , at the size boundary
UPL-06	Corrupted / truncated image is handled gracefully	Medium	expected	corrupted , corrupt image , truncated file , malformed file
UPL-07	Replace / re-upload photo before saving	Low	expected	replace photo , re-upload , change photo

8. Navigation — where the user lands

ID	Scenario	Priority	Grounding	Detection keywords
NAV-01	Success notification / confirmation is shown on save	High	explicit	success notification , success message , confirmation shown
NAV-02	Admin is redirected to the employee list after save	Medium	explicit	redirect , navigates to , employee list
NAV-03	Cancel discards input and returns to the list without saving	Medium	expected	cancel discards , cancel returns , cancel without saving
NAV-04	Navigating away with unsaved changes warns the user	Low	expected	unsaved changes , navigate away warning

9. Search — post-create discoverability

ID	Scenario	Priority	Grounding	Detection keywords
SCH-01	New employee is immediately searchable by employee ID	High	explicit	search by id , searchable by id
SCH-02	New employee is immediately searchable by name	High	explicit	search by name , search-able by name
SCH-03	Partial-name search returns the new employee	Medium	expected	partial name , partial search , partial match
SCH-04	Search is case-insensitive	Medium	expected	case-insensitive search , case insensitive search
SCH-05	New record is findable with no reindex delay	Medium	explicit	immediately searchable , no reindex delay , searchable straight away

10. Data Integrity — what is stored is what was entered

ID	Scenario	Priority	Grounding	Detection keywords
DAT-01	Record is persisted and retrievable after save	High	explicit	persisted , record created , stored , saved to
DAT-02	All entered values are stored exactly (no truncation / mutation)	Medium	expected	stored exactly , no truncation , values preserved , data integrity
DAT-03	Uploaded photo is persisted and retrievable	Low	expected	photo persisted , image stored , photo retrievable
DAT-04	Concurrent creation of the same ID — only one succeeds	Medium	expected	concurrent , simultaneous , race condition

11. Integration — the record does not live alone

ID	Scenario	Priority	Grounding	Detection keywords
INT-01	New employee propagates to dependent views (directory / list)	Medium	expected	propagates , dependent module , appears in directory
INT-02	Success notification is delivered per spec (toast / email)	Low	expected	notification delivered , email notification , toast
INT-03	An audit-log entry is created for the new employee	Low	expected	audit log , audit trail , audit entry

12. Usability — the human using the form

ID	Scenario	Priority	Grounding	Detection keywords
USA-01	Mandatory fields are visually marked	Low	expected	mandatory field marked , re-quired field indicator , marked as re-quired
USA-02	Validation errors are clear and actionable	Medium	expected	clear error message , actionable error , helpful validation
USA-03	Form is keyboard-navigable with associated labels (a11y)	Low	expected	keyboard navigable , accessibility , associated label , a11y

Scoreboard (filled by the audit harness)

Run `npx ts-node scripts/add-employee-audit.ts` . It generates the current Test Case Lab output for this requirement, checks each scenario above, and prints a per-category ✓/✗ table plus the ordered missing list. The number that matters is not a percentage — it is “what would a senior QA notice is missing,” worst-priority first.